

Point-in-Time Multi-Agent LLM Trading: A Reproducible LangGraph Stack with Regime-Conditioned Case Study Evaluation and Fair Backtesting

Rohit Edward Martires

Independent researcher, Goa, India

`rohitmartires14@gmail.com`

29 May 2026

Abstract

Published LLM trading stacks are often hard to reproduce, rarely enforce temporal data boundaries, and seldom report net returns after fees. We present **Hindsight 20/20**, an open **LangGraph** stack with structured **Pydantic** outputs, point-in-time data access, optional **ticker anonymization**, and a fee-aware **paper backtest**. The primary contribution is reproducible systems engineering with frozen CSV artifacts—not a universe-scale alpha benchmark. As an illustrative case study, we evaluate the **full** pipeline on **RELIANCE.NS** and **TCS.NS** in **four pre-specified regime windows** (two tickers $\times$ exogenous bear/bull labels; **one trajectory per window**, no seed averaging). In both bear windows, the agent loses less than buy-and-hold with lower drawdown; bull outcomes diverge (RELIANCE lags B&H; TCS nearly tracks it). Heterogeneous bull outcomes are consistent with a structurally defensive posture;

whether this reflects coordinated de-risking or reduced equity exposure is underdetermined at $N=1$. Code: <https://github.com/RMartires/hindsight>

Keywords: algorithmic trading; large language models; multi-agent systems; reproducibility; point-in-time data; market regimes.

0.1 Introduction

Researchers are experimenting with LLMs inside trading workflows amid rapid growth in LLM-driven agent systems (Bădică et al., 2025), but rarely report how the **same stack** behaves across **different underlying trends**, enforce point-in-time data at the adapter layer, or report **net** returns after stated fees.

What this paper is. **Hindsight 20/20** extends **TradingAgents** (Xiao et al., 2024) with schema-constrained outputs, a simulation end date, optional ticker anonymization, and a scriptable fee-aware backtest tied to **frozen CSV artifacts**.

What this paper is not. We do **not** claim state-of-the-art risk-adjusted returns, regime prediction, or generalization from a broad universe. Each regime window is **N=1** (no multi-seed variance). Frontier benchmarks (Jeon and Lee, 2026; Miyazaki et al., 2026) average many seeds across index universes; our scope is narrower and labeled as such (Section 4.1).

We evaluate the **full** pipeline on **RELIANCE.NS** and **TCS.NS** in **four pre-specified regime windows** (bear and bull per ticker), each from \$100,000 paper cash.

0.2 Related Work

LLM agents and classic MAS. Multi-agent trading frameworks orchestrate specialist roles (Xiao et al., 2024; Yu et al., 2023; Yang et al., 2023). Bădică et al. (2025) contrast emergent LLM agents with classic BDI multi-agent systems. **TradingAgents** supplies our graph skeleton; **AlphaAgents** (Zhao et al., 2025) validates institution-style role separation;

QuantAgents (Li et al., 2025) emphasizes manager-led meetings—comparators for our debate loop, but neither foregrounds adapter-level point-in-time routing or frozen schedule CSVs.

Methodological standards and bias control. Nguyen and Pham (2026) propose Consolidated Minimum Standards for LLM financial multi-agent evaluation—contamination control, net-of-cost returns, and regime awareness—and introduce the Coordination Breakeven Spread (CBS). Bailey et al. (2014) warn that undisclosed configuration search makes strong backtests easy to overfit; we respond with pre-specified windows and frozen artifacts. While TradingAgents provides the foundational graph, Hindsight 20/20 implements the point-in-time policy and net-cost ledger those standards recommend.

Anonymization and fine-grained tasks. Jeon and Lee (2026) (*BlindTrade*) show that identifiers such as AAPL can trigger memorized pretraining rather than active reasoning; they anonymize S&P~500 names and report Sharpe $1.40, \pm 0.22$ across 20 seeds. Hindsight’s optional STOCK_#### aliases pursue the same goal in a LangGraph replication stack. Miyazaki et al. (2026) prove fine-grained analyst sub-tasks outperform coarse prompts; our analysts are tool-bound to indicators and statements (Section~3.2). Dewansh (2026) defines *explainable stability* and rationale persistence; our Pydantic stage outputs support audit trails from outlook to final_signal.

Benchmarks and simulation realism. Qian et al. (2026) (*Agent Market Arena*) show agent architecture often matters more than LLM backbone—motivating our fixed-model, regime-varying case study. Papadakis et al. (2025) (*StockSim*) define order-level simulation with latency and slippage; our candlestick ledger with stated fees is deliberately simpler, with StockSim-class realism as future work.

Scope contrast. Jeon and Lee (2026), Li et al. (2025), and Miyazaki et al. (2026) stress multi-seed index evaluation; Hindsight stresses reproducible temporal integrity on four pre-specified windows (Table 2).

0.3 Method / System

TradingAgentsGraph: analysts $\rightarrow$ bull/bear debate $\rightarrow$ trader $\rightarrow$ risk. Section 4 uses `PAPER_ABLATION=full. backtest_mvp.py --dates-csv` produces wide schedule CSVs.

0.3.1 Analyst task decomposition

Stage	Tools	Tasks
Market	<code>get_stock_data</code> , <code>get_indicators</code>	OHLCV + up to 8 Tier-1 indicators (SMA, EMA, MACD, RSI, Bollinger, ATR, VWMA, MFI)
Fundamentals	<code>get_fundamentals</code> , statements	Revenue, margins, balance-sheet trends
News / Sentiment	<code>get_news</code> , <code>get_global_news</code>	Company and macro news synthesis
Debate $\rightarrow$ Risk	—	Structured stances $\rightarrow$ BUY/HOLD/SELL

0.3.2 Ticker anonymization

Deterministic `STOCK_####` aliases for LLM I/O; vendor round-trip de-anonymization; news proper-noun scrubbing. Frozen E1: `enable_anonymization=True`.

0.3.3 Backtest metrics

We emphasize **net return**, **MDD**, **fee drag**, and **alpha vs buy-and-hold**. **Excess Sharpe** uses daily equity from frozen CSVs with a **6.5% p.a.** risk-free rate (India short-rate proxy)¹. Sortino and Calmar are secondary on short windows (67–111 days).

¹Annualization: $\sqrt{252}$; NSE $\approx$ 240–250 trading days.

0.4 Case Study Evaluation

0.4.1 Scope and limits (Section 4.1)

Two liquid NSE large-caps; **four regime windows**; **N=1 trajectory per window**; exogenous bear/bull labels ($\pm 10\%$ underlying B&H); pre-specified calendars in **regime-timeranges**. Not a hypothesis test about expected alpha.

0.4.2 Positioning (Table 2)

	Hindsight	BlindTrade	QuantAgents	Expert Teams
Primary goal	PIT reproducibility	Anonymized policy	Meetings	Hierarchy
Seeds	N=1/window	20	Multi-year	50
Universe	2 NSE names	S&P 500	Simulation	TOPIX 100
Costs	Net (Zerodha)	Varies	Varies	Varies

Table 2: Qualitative comparison; Hindsight is not comparable on seeds/universe.

0.4.3 Results

Table 3. Primary metrics from frozen CSVs (\$100,000 start).

Ticker	Regime	Days	B&H	Agent	Gross	Fee	α	Ag. MDD	B&H MDD
RELIANCE	Bear	111	-17.4%	-7.5%	-6.5%	1.0pp	+9.9pp	5.1%	21.0%
RELIANCE	Bull	108	+15.1%	+5.6%	+6.2%	0.6pp	-9.5pp	2.8%	11.0%
TCS	Bear	88	-20.4%	-4.8%	-2.9%	1.9pp	+15.6pp	6.8%	23.9%
TCS	Bull	67	+21.9%	+19.6%	+21.8%	2.2pp	-2.3pp	1.7%	5.5%

Table 3: Net return, gross return, fee drag, alpha vs B&H, and MDD.

Table 4. Excess Sharpe (6.5% p.a. R_f), Sortino, Calmar (post-hoc on frozen equity).

Ticker	Regime	Excess Sh. (agent)	Excess Sh. (B&H)	Sortino	Calmar
RELIANCE	Bear	-4.27	-2.51	-3.10	-0.96
RELIANCE	Bull	+0.74	+1.25	+3.67	+1.94
TCS	Bear	-2.69	-3.13	-2.25	-0.71
TCS	Bull	+3.86	+3.27	+24.22	+11.88

Table 4: Secondary risk-adjusted metrics; short windows make ratios noisy.

Patterns. Bear (2/2): the agent generated smaller losses and lower MDD than B&H; SELL-heavy signals in both windows. Bull: TCS nearly tracks B&H; RELIANCE generated a positive return (+5.6%) but lagged by about 9.5pp; fee drag (about 0.6pp) is insufficient to explain the gap; 74 SELL days in the bull window is the plausible binding factor. Internal bull stances can remain buy-leaning while final signals stay defensive.

Capture and exposure. Approximate upside/downside capture ratios reveal a ticker-level distinction. RELIANCE exhibits 43% downside capture and 37% upside capture, a near-symmetric reduction consistent with effective equity exposure below 1.0 rather than selective de-risking. TCS exhibits 24% downside capture and 89% upside capture, a meaningfully asymmetric profile. At N=1 these ratios are descriptive, not diagnostic, but they counsel caution in attributing RELIANCE bear outperformance to regime-aware coordination; a structurally low-exposure posture produces the same pattern mechanically.

Ticker-vs-regime ambiguity. The secondary risk-adjusted metrics (Table 4) reorganise the story: the agent exceeds B&H excess Sharpe in both TCS windows and trails in both RELIANCE windows, a ticker-level split. This reading is equally consistent with the four data points as the regime framing. Neither interpretation is statistically supportable at N=4; both are noted for transparency.

RELIANCE bear Sharpe note. In the RELIANCE bear window the agent records a smaller absolute loss (-7.5% vs -17.4%) yet a worse excess Sharpe (-4.27 vs -2.51 for B&H). A near-cash, low-volatility posture with a small negative excess return produces a large negative annualised Sharpe on a short window; this is a mechanical artefact of the ratio on concentrated SELL sequences, not a contradiction in the data.

0.4.4 Figures

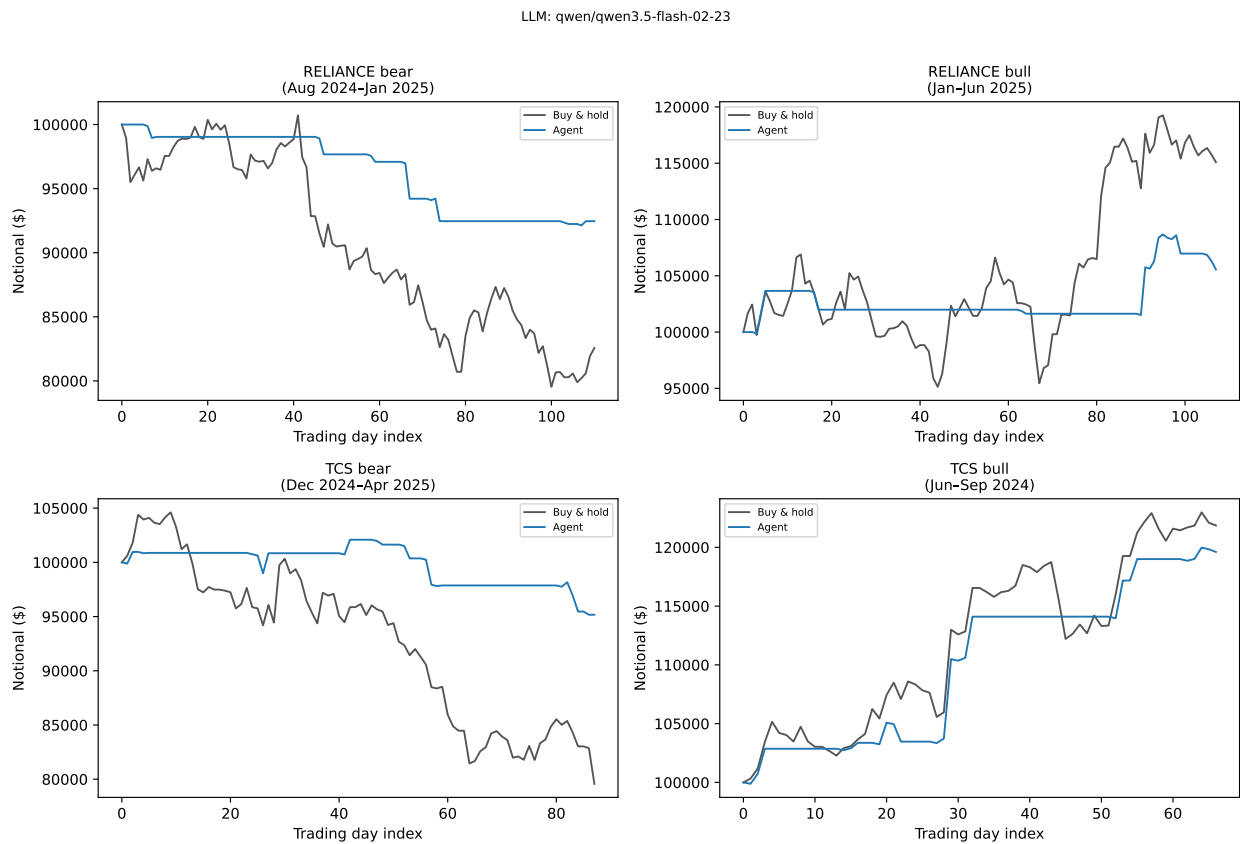


Figure 1: Agent vs buy-and-hold equity (2×2 grid).

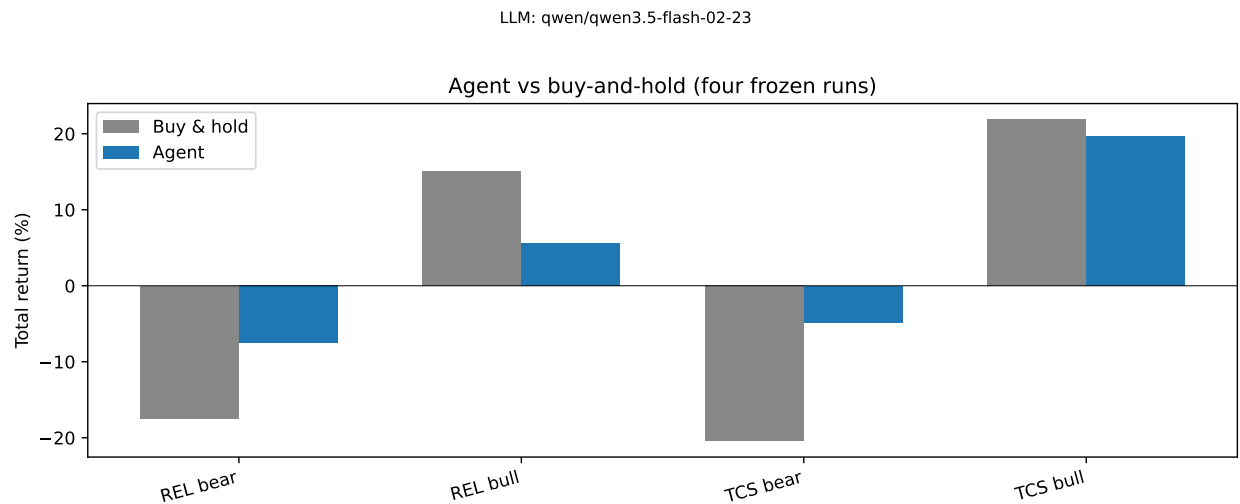


Figure 2: Total returns (four windows).

LLM: qwen/qwen3.5-flash-02-23

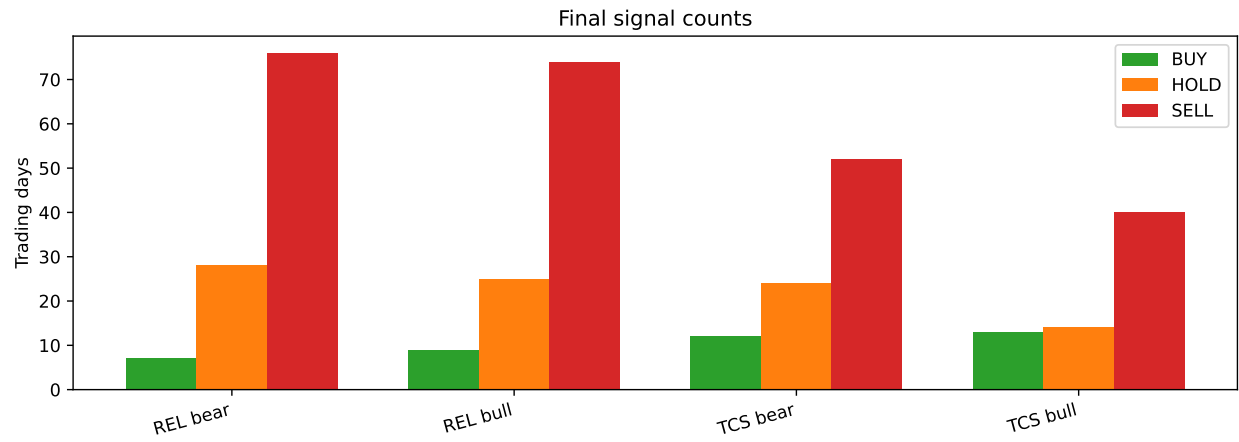


Figure 3: Signal counts.

LLM: qwen/qwen3.5-flash-02-23

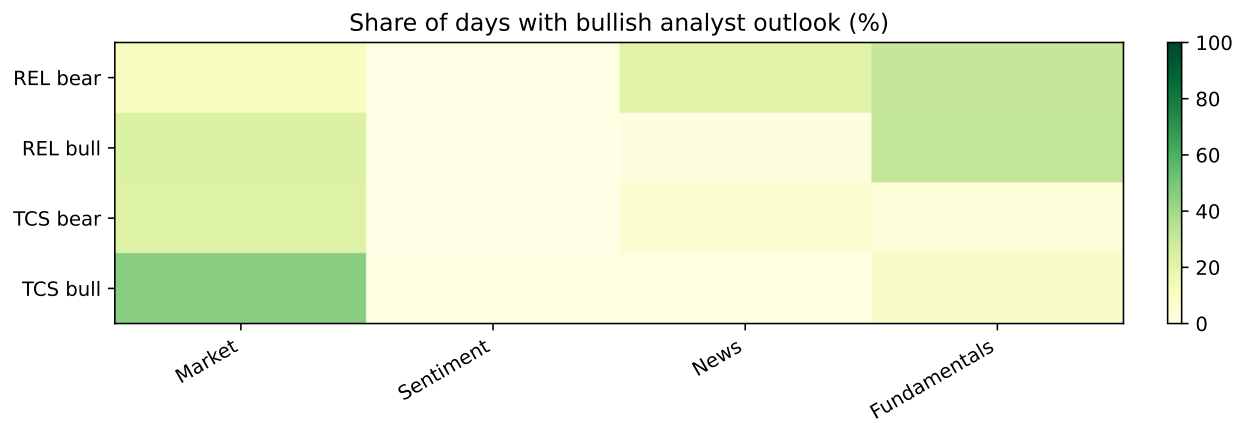


Figure 4: Bullish analyst-outlook share.

0.5 Conclusion

We presented a reproducible point-in-time LangGraph stack and a **scoped four-window case study**. Bear windows show lower loss and drawdown than B&H; bull outcomes are heterogeneous. Risk-adjusted metrics split by ticker rather than regime—both framings are equally consistent with N=4 data points, and no causal claim about coordination quality is made.

Limitations: N=1 per window; two tickers; one LLM; anonymization not ablated; no multi-seed or index-universe evaluation; no latency CBS. **Future work:** seeds, broader PIT universes, anonymization ablations.

0.6 Contributions

1. Reproducible backtest harness and frozen CSV artifacts.
2. Point-in-time policy and optional anonymization (ablation deferred to future work).
3. Structured Pydantic stage outputs.
4. Multi-agent graph with tool-bound analyst decomposition.
5. Honest regime-conditioned case study with documented limits.

Acknowledgements

The agent graph implementation builds on **TradingAgents** (Xiao et al., 2024; Tauric Research, Apache License 2.0).

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